# Final Project Progress Report

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## Introduction

Law enforcement agencies use a variety of technologies to monitor and collect information on the public. These range from aerial devices, to body cameras, to police cars which automatically scan and capture license plate numbers that come into view. They capture location, date, times, sound, and pictures sending all this information into a central server. Government officials are constantly promoting these different technologies for public safety, but what are the possible consequences and threats to our privacy? Some questions we have are what types of technologies are being used, which agencies are using them, and what type of technologies is each type of law enforcement likely to use?

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## Data

***Dataset 1: Atlas of Surveillance***

A dataset from the Electronic Frontier Foundation which represents the many different U.S. law-enforcement agencies and what surveillance technologies they use, such as body cams, drones, ALPRs, etc. The unit of observation would be a single technology used by an agency and each column measures the details of the agency and Technology.

***Dataset 2: Automated License Plate Reader Dataset, from The Electronic Frontier Foundation***

This dataset is from the Electronic Frontier Foundation and the unit of observation is agencies which use ALPR data. The data seems to from 2016-2017 and seems to tally how many cars have been scanned by ALPR and how many of the vehicles were deemed “hot,” either because they were stolen, or the owner has an outstanding warrant, etc. The dataset also include links to data sharing details and several with reports and ratios based on the data in the table. The data here needed to be cleaned.

For the states column, all states were formatted to be capitalized and abbreviated, while cells with null or empty values were given the place holder name “Not provided.”

***Dataset 3:Atlas of Surveillance: Southwestern Border Communities***

A dataset of Southwestern border communities and the types of surveillance technologies they use. The data was collected with the help of the Electronic Frontier Foundation and University of Nevada Reno’s Reynolds School of Journalism. The unit of observation are more agencies from border states, such as California or Texas, and what types of technology they have been reported using. Blank county entries from this dataset were filled with Census area if they did not have an actual county. Multiple counties listed in one cell were also separated.

***Dataset 4: Scholars Under Surveillances***

This dataset is from March of 2021 and is from a report of a student from the University of Nevada, Reno. The unit of observation are agencies from colleges and what types of technology they have been reported using. Many of the cells and columns contain null or empty values which still need to be addressed and cleaned.

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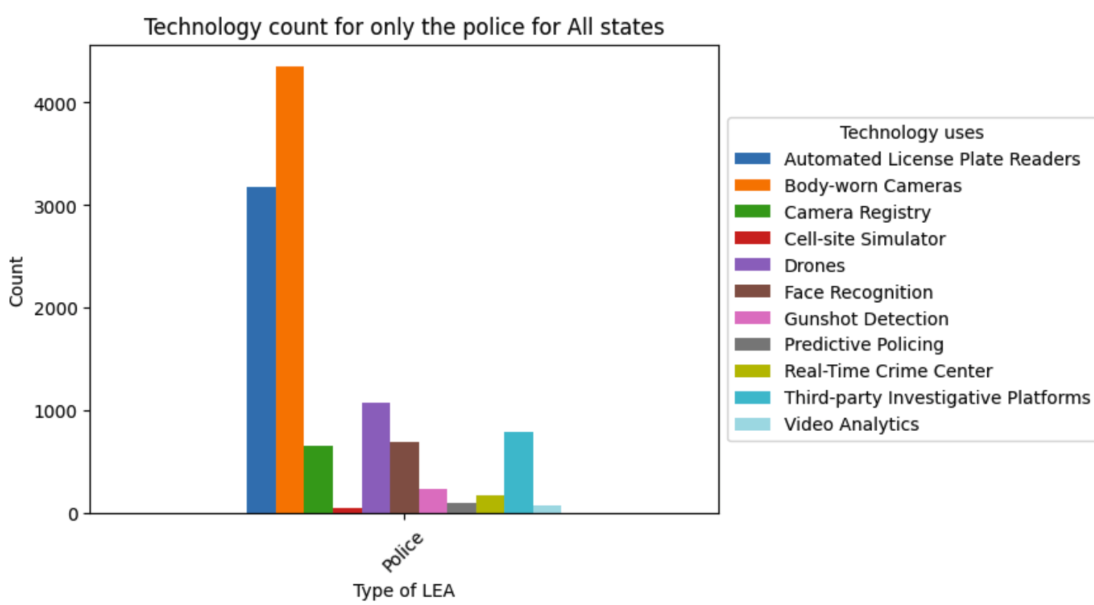
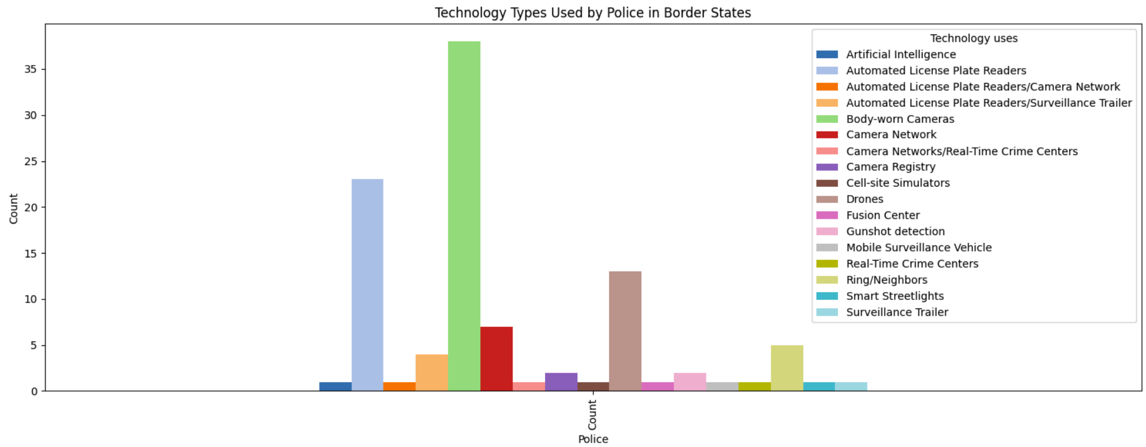
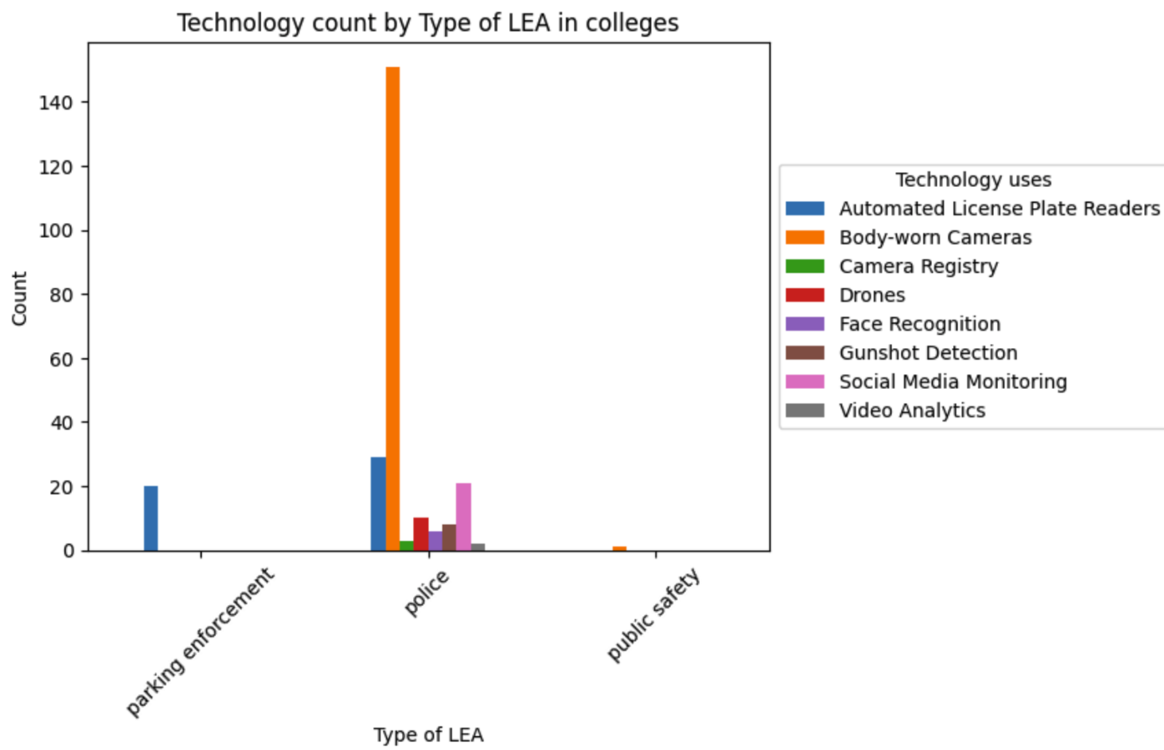
# EDA






**Technology count by Type of Law enforcement Agencies for All states, colleges, and border states:**

The graphs above all depict data from the police of each institution (All U.S. states, U.S. colleges, and border states [CA, AZ, TX, NM]) as well as the amount of each reported technology type the police have been reported using. The top two technologies which all three graphs displayed was Body cam then ALPR usage. The biggest difference between all three graphs would be the third highest technology use from the college's graph, which was social media monitoring. But generally, all three graphs have generally the same shape. The reason why the police department was chosen to be evaluated, compared to any other law enforcement, was because each dataset had the most information from the police department compared to other Law enforcement agencies. The second law enforcement agency with the most data was the Sheriff Department.

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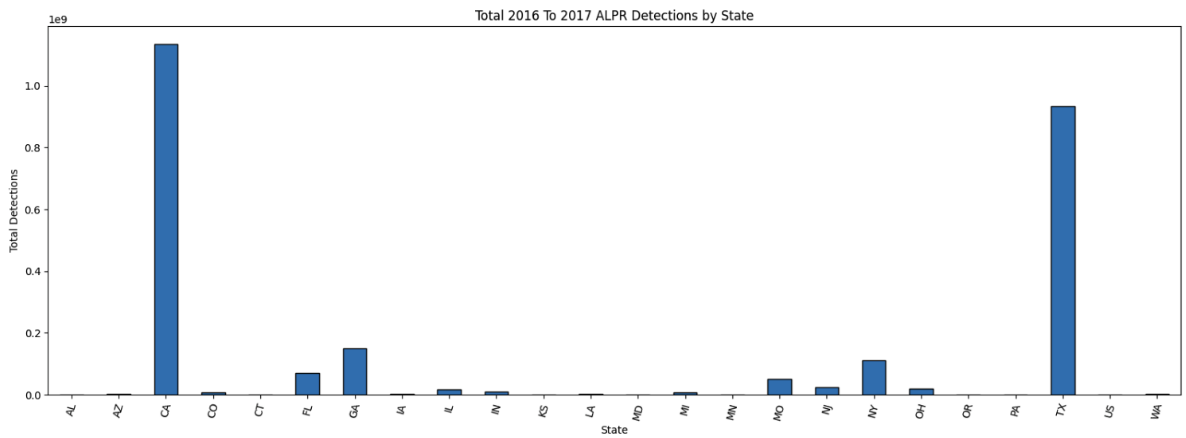


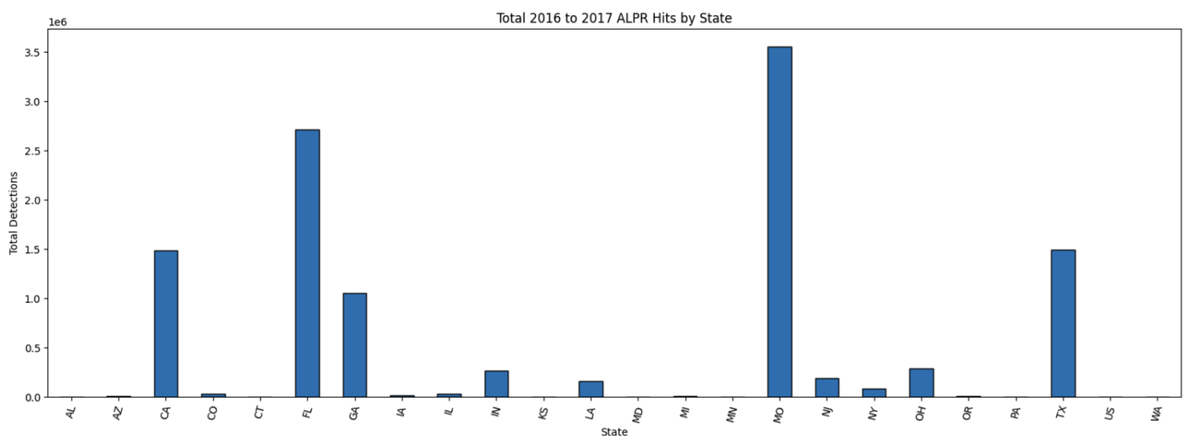


**Total ALPR Detections and Hits from 2016 to 2017 by states:**

The two graphs above show the amount of the license plates an Automated License plate reader scans from the years 2016 to 2017 and is grouped by states. The top graph shows the total amount of cars that were scanned, while the second shows the amount of cars that were deemed “hot” due to either being stolen, the registration being expired, or even tracking vehicles involved in drug trafficking. The two states with the most plate scans are California and Texas, which makes sense in terms of population. The top two states from the hits graph are surprisingly Missouri and Florida. This could possibly be due to these states having provided more, non-empty, information on hits. According to the graphs, out of the cars detected in California, only 0.13% of cars were deemed “hot” in comparison to the millions of cars scans.

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| | Ref | Agency | City | County | State | TypeofLEA | TypeofJuris | Technology | Vendor | Summary |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | EDU016 | California State University San Marcos Police ... | San Marcos | San Diego County | CA | police | University | Body-worn Cameras | Axon | California State University San Marcos Police ... |
| 1 | EDU112 | Palomar College Police Department | San Marcos | San Diego County | CA | police | University | Automated License Plate Readers | None | The Palomar College Police Department began us... |
| 2 | EDU124 | San Diego State University Police Department | San Diego | San Diego County | CA | police | University | Automated License Plate Readers | None | The San Diego State University Police Departme... |
| 3 | EDU125 | San Diego State University Police Department | San Diego | San Diego County | CA | police | University | Body-worn Cameras | None | The San Diego State University Police Departme... |
| 4 | EDU163 | University of California San Diego Police Depa... | La Jolla | San Diego County | CA | police | University | Body-worn Cameras | None | As of 2020, all University of California, San ... |
| 5 | EDU164 | University of California San Diego Police Depa... | San Diego | San Diego County | CA | police | University | Gunshot Detection | ShotSpotter | The University of California San Diego Police ... |
| 6 | EDU165 | University of California San Diego Transportat... | San Diego | San Diego County | CA | parking enforcement | University | Automated License Plate Readers | Park Assist | The University of California San Diego has use... |

**Data specifically about our school:**

We wanted to see specifically what kind of surveillance goes on at our school from our dataset, since it directly affects us. The table above was specifically extracted using SQL and gives us a table of all schools from our college dataset that are from San Diego county. SDSU specifically is only, from our dataset, surveillance by our police, which uses body cam footage and ALPRs.

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## Analysis

We have chosen a classification model to predict the Technologies used by a type of Law Enforcement Agency using City, State, Type Of Law Enforcement Agency, and Jurisdiction as the features. The dataset was cleaned, split into training and testing groups, and then was implemented with Random forests. To check the correctness of our model, we used F1 scores to see how close our results from the model were to what they should have been. Most of our results resulted in as 1.0. We were successfully able to calculate the probability that each Technology would be used by a type of Law Enforcement Agency.

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## Difficulties

Some difficulties we've encountered would be seeing if we need to use all of our datasets. Some of the datasets repeat information with each other, so it was difficult to see what columns we need to remove or keep. All our dataset, except the ALPR dataset, contained the same 5 columns: Agency, City, State, TypeOfLEA, TypeOfJursidiction, And Technology. So we combined all our datasets into one dataframe based on their values

There was one dataset which contained information we were really interested in, but it was not organized in a way that complemented the rest of the data we had. The Automated License Plate Reader dataset also has information on data sharing, which we considered analyzing, but many of the links to the data sharing information led to documents that had been converted to images. Manually extracting the data would have taken more time than given for the project, and attempting to extract the text using Computer Vision would have required another project and is beyond the scope of this class. Still, that would be an interesting exploration.

We also found that the datasets were inconsistent with how they collected the county. One dataset was one county per record, while another listed multiple counties in a record. We were not able to split those rows into counties without either duplicating data or guessing at how the data split by the counties listed, so we had to shift our analysis to the city level instead.

Cleaning our data was one of our difficulties. Numeric values took a very long time to clean. Our ALPR dataset needed to be stripped of extra spaces and characters, which seemed to always reappear. The values then needed to be converted into numeric values. Many of our datasets were filled with typos that also needed to be evaluated and cleaned, slowing down our progress.

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## Summary

Our project investigates how law enforcement agencies across the U.S use surveillance technologies, as well as what it means for public privacy. Using our four datasets, we were able to clean our data, which includes fixing inconsistent county labels, handling missing values, splitting multi-county entries, and converting numeric fields into usable data. In our graphs and visualizations, we found that the law enforcement agency which surveillance the most civilians was the police departments across the country. The police used body cam and automated license plate readers to surveillance their communities the most. We used visualization and SQL to find trends and patterns in our datasets, including data from SDSU. Using Random Forests for our classification model, we calculated the probability that each Technology would be used by a type of Law Enforcement Agency. We used F1 scores to check the precision and recall of our model. Some challenges we faced included duplicated or inconsistent data across datasets, numeric fields containing mixed characters; missing or hard-to-extract data, and deciding which datasets were necessary or redundant. Despite these issues, we have created a consistent final project which answers our questions of: what types of technologies are being used for surveillance, which agencies are using these technologies, and what type of technologies is each type of law enforcement agency likely to use?

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